1. Examine different microscopy techniques used in nanomaterial analysis and compare their working principles.
2. Discuss the classification of nanostructures into zero-, one-, two-, and three-dimensional systems, providing suitable examples for each category.
3. Analyze the advantages and limitations of different nanomaterial synthesis techniques such as Sol–Gel, CVD, PVD, and co-precipitation.
4. Examine the structure, properties, and applications of carbon nanotubes and graphene in modern nanotechnology.
5. Examine the differences between MRAM, DRAM, and FRAM in terms of structure, operation, and data retention characteristics.
6. Analyze how phase change materials enable reversible data storage, including their structural transformation and applications in modern memory systems.
7. Discuss the multilayer structure of a hard disk drive (HDD) and explain the chemical and functional role of each layer.
8. Compare the layered architecture of magnetic (HDD) and optical (CD/DVD) storage systems, emphasizing material selection and working principles.
9. Discuss the environmental impacts caused by chemical materials used in computing systems and explain how these effects can be mitigated using green and sustainable alternatives.
10. Evaluate the environmental impact of computing systems using the Life Cycle Analysis approach from raw material extraction to end-of-life disposal.
11. Highlight the significance of robotics in future computing and smart electronic system management.
12. Discuss any two futuristic innovations in computing systems, including carbon-neutral computing, and explain their importance in sustainable technology development.